

Machine Learning Lab – Final Term Project Proposal

Project Title

Loan Default Detection Using Machine Learning

Problem Statement

Banks and financial institutions face financial losses when customers fail to repay loans. Manually reviewing loan applications is time-consuming and may lead to inaccurate decisions. This project aims to use machine learning to predict whether a loan applicant is likely to default based on financial and personal information.

Dataset Source

Dataset will be taken from one of many on Kaggle.

Target Prediction / Objective

The objective of this project is to predict loan approval/default status using applicant information such as income, credit history, education, and loan amount.

Target Variable:

- **Loan_Status**
 - Y = Approved
 - N = Not Approved / Default Risk

Proposed ML Models

The following machine learning models will be used and compared:

1. Decision Tree Classifier
2. K-Nearest Neighbors (KNN)

Proposed Workflow

- Data cleaning and preprocessing
- Handling missing values
- Encoding categorical data
- Feature engineering
- Training and testing ML models
- Model comparison using evaluation metrics
- Data visualization and interpretation

Evaluation Metrics

Models will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Visualization

At least one meaningful visualization will be included, such as:

- Loan approval distribution
- Credit history vs loan status
- Confusion matrix heatmap

Tools & Technologies

- Python
- Jupyter Notebook
- pandas
- NumPy
- scikit-learn
- matplotlib
- seaborn

Group Members

- Member 1: Zarak Khan
- Member 2: Anas Rehman